

ECSE 552 Project Proposal

Teaching a Robot Policy a New Object from Action-Free Videos:
Object-Invariant Inverse Dynamics for Pseudo-Labeling

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1 Title and Summary

- **Working title:** Teaching a Robot Policy a New Object from Action-Free Videos: Object-Invariant Inverse Dynamics for Pseudo-Labeling.
- **One-line summary:** use action-labeled demos of one object plus action-free videos of a second object to train a policy for the second object, entirely in simulation.
- **Central question:** where should domain invariance be applied — in the model that labels the videos, in the policy, or both?

2 Background

2.1 Application (A1)

- Learning robot manipulation skills (picking up and handing over objects) from demonstrations.
- Imitation learning / behavior cloning: a neural network maps camera images and joint positions to motor commands.
- Current state of the art for this setting: ACT (Action Chunking with Transformers) [1].

2.2 Problem (A2)

- Behavior cloning needs demonstrations where every frame is paired with the motor command used.
- These labeled demos are slow and expensive to collect (a human must teleoperate the robot).
- Videos of a robot doing a task are much cheaper, but contain no motor commands.
- A policy trained on one object fails on a new object that needs a different grasp.
- **Question:** can action-free videos of a new object, plus labeled demos of a known object, train a policy for the new object?

2.3 Why a Solution Is Needed (A2)

- Collecting labeled demos for every new object does not scale.
- Prior domain adaptation work [2] assumes the task is the same across domains and only appearance changes (e.g. the color of the object).
- Here the correct actions change with the object, so standard adversarial alignment may remove information the policy needs.

- Pseudo-labeling videos with an inverse dynamics model works when the labeler generalizes; whether it survives an object change is untested.

2.4 Constraints (A3)

- **No target action labels:** only image frames of the new object may be used for training.
- **Closed-loop task success:** the goal is a high success rate when the policy controls the simulated robot.
- **Real-time inference:** the policy must output actions within the 50 Hz control budget (< 20 ms per step).
- **Data efficiency:** at most ~ 50 labeled source demonstrations.
- **Selective invariance:** the labeling model should ignore object identity; the policy must stay sensitive to it.
- **Robustness to compounding error:** small action errors must not drift the robot off the training distribution.

3 Aims and Goals

3.1 Aims

- Aim 1:** Build an inverse dynamics model (IDM) that infers motor commands from consecutive frames and stays accurate when the object changes.
- Aim 2:** Use it to pseudo-label action-free videos of a new object, then train an ACT policy on labeled + pseudo-labeled data.
- Aim 3:** Test where invariance belongs: object-invariant IDM vs. object-invariant policy vs. both.

3.2 Scope (A4)

- **In scope:** one source task (ALOHA Transfer Cube), one target object (elongated bar at random rotation), simulation only.
- **In scope:** evaluation of the full pipeline in closed-loop simulated rollouts.
- **Out of scope:** real robots, human videos, other robot arms, multiple tasks, language instructions.

3.3 Proposed Solution (A5)

- **Key insight:** in ALOHA an action is the next target joint position, so the action is visible in the next frame.
- An IDM reading frames t and $t + 1$ is mostly estimating arm pose, which does not depend on the object.
- Therefore, make the IDM object-invariant with an adversarial domain classifier and a gradient reversal layer.
- Do *not* make the policy invariant, since it must grasp each object differently.
- **Pipeline:** train IDM on cube data $\rightarrow$ label bar videos $\rightarrow$ train ACT on both.

4 Methodology

4.1 Data Generation (A6)

- **Simulator:** ALOHA bimanual environment in MuJoCo [3], via LeRobot [4] `gym-aloha` and the ACT repository.
- **Source data:** ~ 50 scripted Transfer Cube demos with images and actions.
- **Target data:** 10–100 scripted demos with a new elongated bar; the scripted policy is extended to rotate the gripper.
- Target actions are hidden from all models; they are kept only for evaluation and for the oracle.
- **Held-out test set:** new random object positions, never used for training.

4.2 Models and Implementation (A5, A6)

- **Policy:** ACT — ResNet-18 image encoder, transformer encoder–decoder, CVAE objective, action chunks.
- **IDM:** ResNet-18 encoder on frames $(t, t+1) \rightarrow \text{MLP} \rightarrow 14\text{-dim action}$; trained with L1 loss on cube data.
- **Object-invariant IDM:** add a domain classifier on IDM features (cube vs. bar frames) with a gradient reversal layer.
- **Training loss (IDM):**

$$\mathcal{L}_{\text{IDM}} = \mathcal{L}_1^{\text{action}} + \lambda \mathcal{L}_{\text{adv}}^{\text{domain}},$$

with λ tuned on a validation split.

- **Final policy:** ACT trained on cube demos + pseudo-labeled bar demos.
- **Framework:** PyTorch + LeRobot; hyperparameters tuned on a validation set of held-out episodes.

4.3 Evaluation (A7)

- **Primary metric:** closed-loop success rate on the bar, 50 rollouts $\times$ 3 seeds per method.
- Subtask success: grasp, lift, handover.
- Pseudo-label quality: IDM action error vs. hidden true actions.
- Inference latency per step (checks the real-time constraint).
- Scaling study: success vs. number of target videos (10, 25, 50, 100).
- Ablation: invariance in IDM only / policy only / both.
- t-SNE plots of cube vs. bar features in the IDM and in the policy.

4.4 Alternative Solutions, All Run on Our Data (A8)

- **Source-only ACT:** trained on cube data only (lower bound).
- **ACT + DANN:** adversarial invariance on the policy encoder, using unlabeled bar frames.
- **IDM pseudo-labeling without adaptation:** VPT / BCO-style.
- **Diffusion Policy:** alternative state-of-the-art policy backbone, trained with the best pipeline.
- **Oracle ACT:** trained on true bar actions (upper bound).

4.5 Why Ours Could Be Superior (A8)

- DANN on the policy forces cube and bar features together, discarding the grasp information the bar needs.

- An unadapted IDM may mislabel frames when it sees an unfamiliar object.
- Our method applies invariance only where the input–output mapping is shared across objects.

5 Feasibility and Resources

- **No robot needed:** all data generation and evaluation run in the MuJoCo simulator, which is open-source.
- **Dataset:** self-generated with scripted policies; ground-truth actions are available for every frame, so pseudo-labels can be scored exactly.
- **Code:** open-source ACT repository (scripted policies, episode recording) and LeRobot (ACT, Diffusion Policy, `gym-aloha`).
- **Compute:** ACT (~ 80 M parameters) trains in ~ 5 h on one RTX 2080 Ti in the original paper [1].
- Estimated ~ 15 – 20 training runs $\rightarrow$ reduce cost with 1–2 cameras instead of 4, lower image resolution, and fewer training steps.
- **GPU source:** [\[TO CONFIRM — university cluster / Colab Pro / personal GPU\]](#).
- **Early check:** confirm the scripted cube policy runs and the new bar object loads before finalizing the plan.

6 Risk Analysis

Risk	Mitigation / fallback
Adding the bar or extending the scripted policy is harder than expected	Use a visual-only object change (color/texture), or the existing Insertion task as target.
Pseudo-labels too noisy for the policy to learn from	Smooth labels over action chunks; filter low-confidence labels; few-shot variant with 5 truly labeled bar demos.
Even the oracle policy fails on the bar	Simplify the object (shorter bar, smaller rotation range); report per-subtask success.
Adversarial training unstable	Tune λ , warm-up schedule for the domain loss; try MMD or OT alignment as alternatives.
Not enough GPU time	Smaller models, fewer seeds, offline action error as secondary evidence.
Results only work in simulation	State as a limitation; sim-to-real transfer listed as future work.

7 Role of Group Members

Member	Main responsibilities
Brik	[TBD]
Zheng	[TBD]
Myriam	[TBD]
Gonzalo	[TBD]

All members: literature review, proposal, presentation, and report sections for their component.

References

- [1] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bi-manual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023. arXiv:2304.13705.
- [2] Shuo Cheng, Liqian Ma, Zhenyang Chen, Ajay Mandlekar, Caelan Reed Garrett, and Danfei Xu. Generalizable domain adaptation for sim-and-real policy co-training. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [3] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.
- [4] Remi Cadene et al. LeRobot: State-of-the-art machine learning for real-world robotics in PyTorch. GitHub, 2024.